Bubble Plots

2026-04-17

Introduction

A bubble plot is a scatter plot enriched with a third continuous variable encoded by point size. It is a standard idiom in business analytics (“revenue × growth × market share”) and in international development reporting (Hans Rosling’s life-expectancy × income × population). The third dimension comes at a real perceptual cost — humans judge area less accurately than position — so a bubble plot’s size aesthetic should be reserved for variables where rank order and rough magnitude matter more than precise comparison.

Prerequisites

A working knowledge of scatter plots, `ggplot2`’s aesthetic mappings, and the perceptual principles of position vs area encoding.

Theory

A bubble plot maps three continuous variables: x, y, and `size`. The size mapping must use an *area* scale — `scale_size_area()` — so that perceived magnitude (area) is proportional to the data value. The default `scale_size()` maps the data to point radius, which means a doubled data value occupies four times the visual area; this is a long-standing bug-as-feature in many bubble-chart implementations and produces misleading impressions of magnitude. Colour can encode a fourth variable, but four-variable bubble plots tend to overwhelm readers; alternatives include a facet on the categorical fourth variable or an animated time slider for longitudinal data.

Assumptions

All three (or four) variables are continuous (or treated as such). The size variable is non-negative and on a scale where zero means “absent”; mapping size to a centred or signed quantity is undefined.

R Implementation

```r
library(ggplot2)

ggplot(mtcars, aes(wt, mpg, size = hp, colour = factor(cyl))) +
  geom_point(alpha = 0.7) +
  scale_size_area(max_size = 12, name = "Horsepower") +
  scale_colour_brewer(palette = "Set2", name = "Cylinders") +
  theme_minimal()
```

Output & Results

A scatter of weight vs. miles-per-gallon with bubble area proportional to horsepower and colour encoding cylinder count. Heavy, high-horsepower cars cluster in the lower-right corner; light four-cylinder cars cluster in the upper-left. The combination of size and colour makes structural patterns visible that would require a four-panel facet to communicate without bubbles.

Interpretation

A reporting sentence (figure caption): "Bubble plot of vehicle weight vs fuel economy (n = 32); bubble area is proportional to horsepower (`scale_size_area`, max area 12), and colour encodes cylinder count." Always specify that area (not radius) is the encoding; readers familiar with poorly designed bubble charts will otherwise assume the wrong mapping.

Practical Tips

- Always use `scale_size_area()` so the visual area is proportional to the data; the default radius mapping double-counts magnitude.
- Set `max_size` deliberately (typically 8–15) to avoid enormous bubbles dominating the panel and obscuring smaller observations.
- For many overlapping bubbles, lower `alpha` to 0.4–0.6 and prefer a sequential fill so density still reads through the transparency.
- Do not map size to negative or signed values; area is undefined for negatives, and the visual mapping is ambiguous.
- For a fourth dimension, prefer a facet over a colour aesthetic on top of size; combining both fights for the reader's attention.
- For time series of bubbles ("Rosling-style"), use `gganimate::transition_time(year)` rather than a static panel; the temporal dimension is the variable that makes bubble plots compelling.

R Packages Used

`ggplot2` for the underlying scatter, `scale_size_area()` for correct area encoding, `gganimate` for time-animated bubble charts, and `ggrepel` for non-overlapping bubble labels when individual observations need to be named.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- ggplot2 grammar and multi-panel layouts